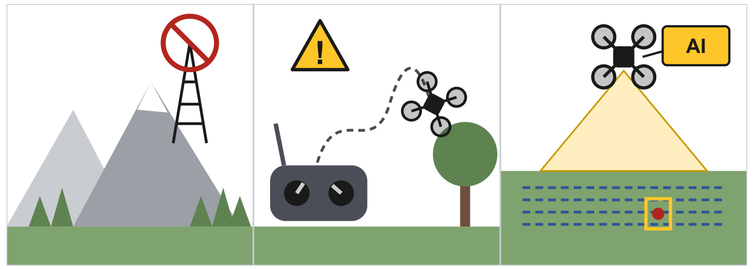
XX-000

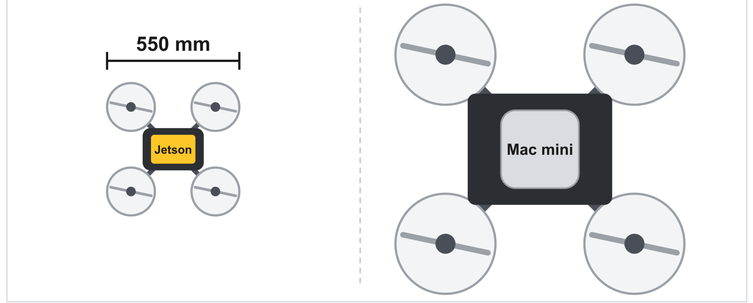
KESTREL: Just Say Who's Missing

Shiva Shrestha

1. The Problem



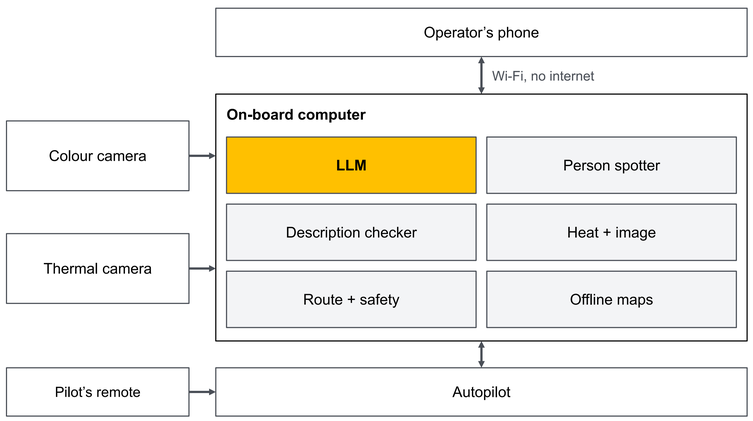
2. Two Drone Models



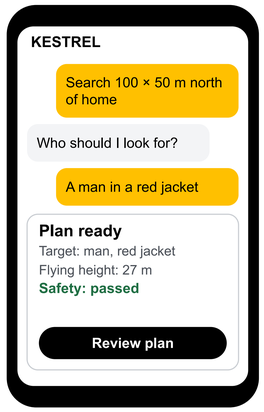
Compact, 550 mm
Jetson · Gemma 4 E2B

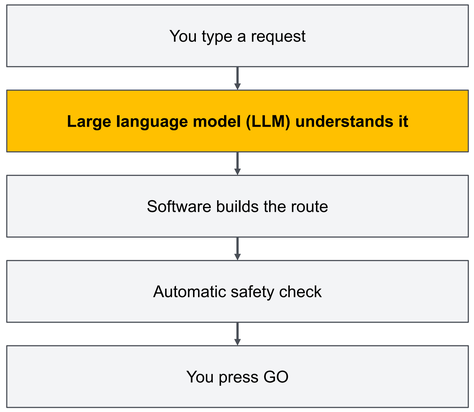
Large, 1.2 m
Mac mini · Qwen3.6-35B-A3B

3. System Overview

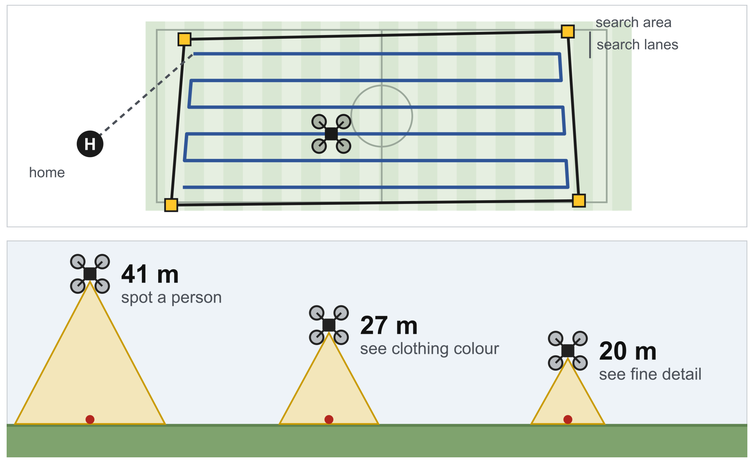


4. Plan a Search by Chatting

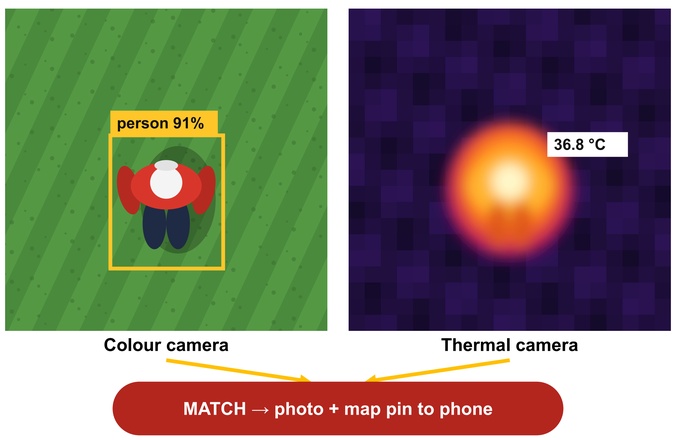




5. Search Pattern and Height



6. Finding the Person

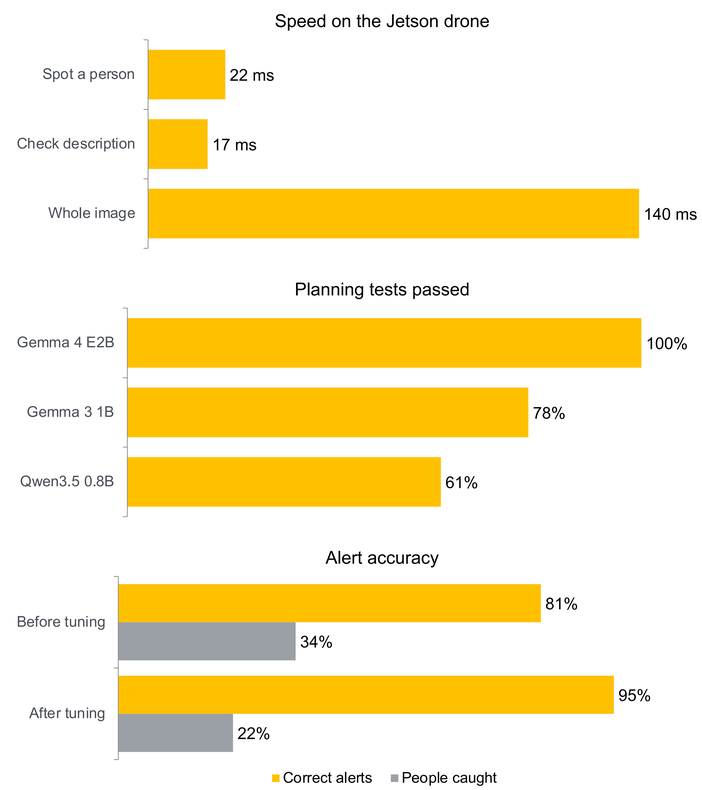


7. Automatic Safety Checks

- ✓ GPS locked
- ✓ Virtual fence
- ✓ Airspace approval
- ✓ Autopilot self-test
- ✓ Cameras and AI
- ✓ Battery level
- ✓ 400 ft limit
- ✓ Pilot's remote
- ✓ Route verified
- ✓ Legal checklist

FAA Part 107 · licensed pilot can take over

8. Results



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